Tom Gannon
Assistant Professor
University of California, Riverside
tom.gannon@ucr.edu

Education and Employment

University of California, Riverside
- Assistant Professor — Fall 2025 - Present

University of California, Los Angeles
- Hedrick Adjunct Assistant Professor — Fall 2022 - Summer 2025
- Supervisor: Raphaël Rouquier

University of Texas at Austin
- Ph.D. - Mathematics — Fall 2016 - Summer 2022
- Advisor: Sam Raskin
- Thesis: *Categorical Representation Theory and the Coarse Quotient*

Michigan State University
- B.A. - Dual Major in Advanced Mathematics and Accounting — Fall 2012 - Spring 2016
- Computer Science Minor

Publications

- Terminal Singularities of the Affine Closure of $T^*(\mathrm{SL}_n/[P,P])$ - Appendix (with N. Grantcharov)
  Submitted.
- Crepant partial resolutions of the nilpotent cone via Hamiltonian reduction (with G. Bellamy)
  Submitted.
- The coordinate ring of the universal centralizer via Demazure operators (with V. Ginzburg)
  Submitted.
- Functoriality of Coulomb branches (with B. Webster)
  Submitted.
- Quantization of the universal centralizer and central D-modules (with V. Ginzburg)
  Submitted.
- The cotangent bundle of G/U_P and Kostant-Whittaker descent
  International Mathematics Research Notices - 2026.
- Differential operators on the base affine space of SL_n and quantized Coulomb branches (with H. Williams)
  International Mathematics Research Notices - 2026.
- Proof of the Ginzburg-Kazhdan conjecture
  Advances in Mathematics - 2024.
- Classification of nondegenerate G-categories
  Submitted.
- The universal category $\mathcal{O}$ and the Gelfand-Graev action
  Selecta Mathematica - To appear.
- The coarse quotient for affine Weyl groups and pseudo-reflection groups
  Transformation Groups - 2025.

Undergraduate Publications
- New results about semipositive matrices (with J. Dorsey, C. Johnson, and M. Turnansky)
  Czechoslovak Mathematical Journal - 2016.
- Linear preservers of semipositive matrices (with J. Dorsey, N. Jacobson, C. Johnson, and M. Turnansky)
  Journal of Linear and Multilinear Algebra - 2016.

Research Seminars and Conferences

- SLMath Workshop: Geometric Representation Theory and 3d Mirror Symmetry — Fall 2026*
- UC Davis Algebraic Geometry — Fall 2026
- SLMath Pathways Workshop: Representation Theory Under the Influence of Quantum Field Theory — Fall 2026
- Coulomb branches and affine quantum groups conference - Centre International de Rencontres Math'ematiques — Summer 2026
- Claremont McKenna Algebra, Number Theory, and Combinatorics — Spring 2026
- UCSB Topology Seminar — Spring 2026
- UC Berkeley Informal String-Math Seminar — Fall 2025
- Northeastern Workshop on Categorification and Symplectic Duality — Summer 2025
- UC Riverside Colloquium — Spring 2025
- Wake Forest Colloquium — Spring 2025

- Oberwolfach Arbeitsgemeinschaften — Spring 2025
- UW Madison Special Session in Geometric and Topological Aspects of Mathematical Physics and Representation Theory — Spring 2025
- Notre Dame Colloquium — Fall 2024
- Perimeter Institute Mathematical Physics — Fall 2024
- University of Chicago Representation Theory — Fall 2024
- MIT Lie Groups — Fall 2024
- UC Riverside Lie Theory — Fall 2024
- University of Georgia Algebra — Fall 2024
- LSU Mathematical Physics and Representation Theory — Fall 2024
- Georgia Southern Poisson Geometry, Diffeology and Singular Spaces — Fall 2024
- University of New Mexico Algebra and Geometry — Fall 2024
- UNC Geometric Methods in Representation Theory — Fall 2024
- Utah State Department Colloquium — Fall 2024
- UC Riverside Special Session on Non-commutative Birational Geometry, Cluster Structures and Canonical Bases — Fall 2024
- UC Riverside Special Session on Non-commutative Algebras in Representation Theory and Topology — Fall 2024
- UC Riverside Special Session in Geometric and Categorical Representation Theory — Fall 2024
- Caltech Algebra — Spring 2024
- Yale Geometry, Symmetry and Physics — Spring 2024
- Northwestern 'Gone Fishing' Poisson Geometry Conference — Spring 2024
- USC Algebra — Spring 2024
- Northeastern Geometry, Physics, and Representation Theory — Spring 2024
- Perimeter Institute Mathematical Physics — Spring 2024
- UW Milwaukee Special Session on Geometric Methods in Representation Theory — Spring 2024
- Southeastern Lie Theory Workshop Series — Spring 2024
- SFSU Special Session on Commutative and Noncommutative Algebra — Spring 2024
- SFSU Special Session on Representations of Lie Algebras and Lie Superalgebras — Spring 2024
- SFSU Special Session on Geometry, Integrability, Symmetry and Physics — Spring 2024
- CMS Special Session on Cluster Algebras in Representation Theory — Fall 2023
- SUNY Buffalo Special Session on Representation Theory and Flag Varieties — Fall 2023
- UC Berkeley Representation Theory and Tensor Categories — Spring 2023
- UCLA Algebra — Spring 2023
- University of New Mexico Algebra and Geometry — Spring 2023
- USC Topology — Spring 2023
- UC Riverside Lie Theory — Spring 2023
- MIT Lie Groups — Spring 2022
- Oberwolfach Arbeitsgemeinschaften — Spring 2022
- Boston College Algebraic Geometry — Spring 2022
- Notre Dame Algebraic Geometry — Spring 2022
- Montana Mathematics — Spring 2022
- Toronto Geometric Representation Theory — Fall 2021
- Academia Sinica Algebraic Geometry — Fall 2021
- Texas A&M Algebraic Geometry — Fall 2020

Honors and Awards

- Simons Foundation Travel Support for Mathematicians ($36,750) — Fall 2026 - Summer 2031
- AMS-Simons Travel Grant ($6000) — Fall 2024 - Summer 2026
- Frank Gerth III Outstanding Dissertation Award — Spring 2022
- Graduate School Dissertation Writing Fellowship — Spring 2022
- Frank Gerth III Teaching Excellence Award — Spring 2019

Teaching Experience

UC Riverside

- Math 007B - Integral Calculus for Life Sciences — Fall 2025
- Math 009B - First Year Calculus: Integration — Winter 2026
- Math 009C - First Year Calculus: Series — Spring 2026
- Math 022 - Calculus for Business — Winter 2026

UCLA

- Math 31 - Differential and Integral Calculus — Fall 2023 (2 sections), Fall 2022
- Math 33 - Linear Algebra and Applications — Fall 2025, Spring 2023, Fall 2022
- Math 115B: Linear Algebra — Winter 2025, Winter 2023

Service

Mentor/Organizer - LSU *Focused Research Community* — Spring 2026

Organizer - UCR *Representation and Lie Theory Seminar* — Spring 2026, Fall 2025

Organizer - AMS Sectional Special Session (Riverside) Fall 2024
Organizer - Los Angeles Workshop on Representations and Geometry (LAWRGe) Summer 2024
Teaching Assistant - U of O Workshop on Algebra and Representation Theory Summer 2023

Math Fellow (Postdoc) Mentorship
- Ruoxi Li Fall 2026 - Summer 2029*

Reading Courses
- Oliver Waite Winter 2026
- Max Steinberg (UCLA) Spring 2023

Peer Review

Referee Work
- *Journal of the European Mathematical Society*
- *Annales Scientifiques de l'Ecole Normale Supérieure*
- *Compositio*
- *Representation Theory* (AMS)
- *Forum of Mathematics, Sigma*
- *Annals of Representation Theory*
- *International Mathematics Research Notices*

Quick Opinion
- *Journal of the European Mathematical Society*
- *Annales Scientifiques de l'Ecole Normale Supérieure*
- *Selecta*
- *Compositio*
- *Proceedings of the London Mathematical Society*
- *Representation Theory*
- *Journal of the London Mathematical Society*
- *Geometric And Functional Analysis*
- *Banff International Research Station*

Outreach

UCLA Math Circle - Deputy Director Fall 2023 - Summer 2025
- Co-director of program for over 400 students in grades K-12 designed to teach accessible introductory topics in theoretical math
- Co-developed new courses for advanced high school students in probability, game theory, and basics of algebraic geometry
- Regularly taught Sunday classes to students in grades K-12
- Administered American Mathematics Competition, Math Kangaroo, AIME, Bay Area Math Olympiad, USAMO

Sunday Morning Math Group - Coordinator Fall 2018 - Fall 2020
- Coordinated logistics related to mathematics outreach program
- Lead marketing campaign to increase awareness of biweekly outreach event
- Boosted attendance from 20 students per session to 100 students per session
- Received 'Featured Event' status on popular Austin website Do512
- Advised mathematics faculty and graduate students on teaching young students
- Administered American Mathematics Competition

UTeach Summer 2020 - Summer 2021
- Co-organized math department efforts to pair graduate student volunteers with local schools in need of additional assistance during COVID-19 pandemic
- Acted as supplemental instructor for an AP Calculus class at Lehman High School
- Organized exhibit for Lehman STEM Fair

Outreach Math Talks Fall 2016 - Present
- Regular speaker at UT Undergraduate Math Club, Sunday Morning Math Group and other locations
- Spoke on panels regarding topics such as applying to graduate school and REUs

Blogging
- How to Run a Math Group - *Notices of the AMS* Fall 2021
- What Is... A Scheme? - *Notices of the AMS* Fall 2017
- Contributor to *AMS Blogs* Spring 2017 - Fall 2018